1.5 Real and Complex Sinusoids

Section 1.4 ended with the equation $y' - ay = e^{ct}$. A particular solution was easy to produce, because we kept e^{ct}. We simply chose the correct multiplier $Y = 1/(c-a)$ in $y_p(t) = Ye^{ct}$. **This section changes the real number c to an imaginary number $i\omega$. The multiplier is now $Y = 1/(i\omega - a)$** . The solution formula $Ye^{i\omega t}$ will stay exactly the same, but we need complex numbers (with real part and imaginary part). The payoff is that we can solve all real problems $y' - ay = A\cos\omega t + B\sin\omega t$ at once.

Many scientific and engineering applications are driven by sources $q(t)$ that oscillate like $\cos\omega t$ and $\sin\omega t$ (**sinusoids**). Pistons go up and down to drive a car, voltages go up and down to drive current (alternating current). The input frequency is ω, and the output frequency is also ω. The problem is to find the *amplitude* and the *phase* in the output (the response to the input). The real solution will be $\boldsymbol{y = M\cos\omega t + N\sin\omega t}$.

This $y(t)$ will be a particular solution (steady solution). It is not the transient solution $y_n(t)$ that decays to zero. *We solve $y' - ay = q(t)$ when the source $q(t)$ is a sinusoid.* For this section and the next, applications come from biology and chemistry and medicine and more. The number a is often a rate constant. It tells the speed of a chemical reaction.

Note that RLC circuits (resistor-inductor-capacitor) produce equations with second derivatives. Those will go into Chapter 2, but RC and RL circuits (first order equations) belong here. Our plan for this section is straightforward: *Real then complex.*

1 (Real) Solve $\boldsymbol{dy/dt - ay = q(t) = A\cos\omega t + B\sin\omega t}$.

This leads to two equations for the two coefficients M, N in $y = M\cos\omega t + N\sin\omega t$.

2 (Complex) Solve $\boldsymbol{dy/dt - ay = q(t) = Re^{i\omega t}}$.

This leads to one easy equation for the coefficient in $y = Ye^{i\omega t}$. But that number Y is complex, so we still have two real numbers to find (real and imaginary parts of Y).

3 (A key idea) **Write the complex number $\boldsymbol{1/(i\omega - a)}$ in its polar form $\boldsymbol{Ge^{-i\alpha}}$.**

The positive number G is the **gain** . The angle α is the **phase lag** . Those have important meanings and they are perfect to graph separately. In many problems (most problems) G and α are more useful than the real and imaginary parts of $1/(i\omega - a)$.

So we need to explain and review complex numbers. They are worth knowing and not difficult. The next page will solve the real problem **1** and the complex problem **2**. We can't simplify the real problem by using cosines alone, because the term dy/dt in the equation would unavoidably involve $\sin\omega t$.

The *Review of the Key Ideas* at the end organizes the important steps.

Real Sinusoids

We want a particular real solution $y(t)$ when the source $q(t)$ oscillates with frequency ω.

First order linear equation	$\dfrac{dy}{dt} - ay = A\cos\omega t + B\sin\omega t.$	(1)

The solution will have the same form $y = M\cos\omega t + N\sin\omega t$ as the source term. By matching the $\cos\omega t$ terms and separately the $\sin\omega t$ terms, you get two equations for M and N. Just subtract $ay = aM\cos\omega t + aN\sin\omega t$ from $dy/dt = -\omega M\sin\omega t + \omega N\cos\omega t$.

$$\frac{dy}{dt} - ay = q \qquad \begin{matrix} \textbf{cos } \omega t \textbf{ terms} & -a\ M + \omega\ N = A \\ \textbf{sin } \omega t \textbf{ terms} & -\omega\ M - a\ N = B \end{matrix} \tag{2}$$

Those two equations tell us M and N in the real solution $y(t) = M\cos\omega t + N\sin\omega t$. I will write down the solution to equation (2), and then describe two ways to find it.

$$\textbf{Source } q = A\cos\omega t + B\sin\omega t \qquad \textbf{Solution } y = M\cos\omega t + N\sin\omega t$$

$$M = -\frac{aA + \omega B}{\omega^2 + a^2} \qquad N = \frac{\omega A - aB}{\omega^2 + a^2} \tag{3}$$

I would find N by eliminating M in equation (2). If you multiply the first equation by ω and the second equation by a, then *subtraction removes* M. The right side is $\omega A - aB$, the left side is $(\omega^2 + a^2)N$. Then N is correct in equation (3). Similarly we find M.

For two equations it is also practical to find M and N from the 2 by 2 **inverse matrix** :

$$\begin{bmatrix} -a & \omega \\ -\omega & -a \end{bmatrix} \begin{bmatrix} M \\ N \end{bmatrix} = \begin{bmatrix} A \\ B \end{bmatrix} \quad \text{gives} \quad \begin{bmatrix} M \\ N \end{bmatrix} = \frac{1}{\omega^2 + a^2} \begin{bmatrix} -a & -\omega \\ \omega & -a \end{bmatrix} \begin{bmatrix} A \\ B \end{bmatrix}.$$

The matrix on the left times its inverse on the right gives the identity matrix I in Chapter 4. That denominator $\omega^2 + a^2$ of the inverse matrix appears in M and N, in the solution (3).

Complex Sinusoid $e^{i\omega t}$

Now we come to the very important input $q(t) = Re^{i\omega t}$. That input is oscillating with frequency ω radians per second. *The output $y(t)$ will oscillate with the same frequency ω.* This is true because a is constant in the differential equation. When $y(t) = Ye^{i\omega t}$ includes the same factor $e^{i\omega t}$, that factor cancels from every term in the equation:

$$\begin{matrix} q(t) = Re^{i\omega t} \\ y(t) = Ye^{i\omega t} \end{matrix} \qquad y' - ay = q \text{ becomes } i\omega Ye^{i\omega t} - aYe^{i\omega t} = Re^{i\omega t}. \tag{4}$$

When we divide by $e^{i\omega t}$, this leaves an easy algebra problem for the complex number Y :

$$\textbf{Response } Y(\omega) \qquad i\omega Y - aY = R \quad \text{gives} \quad Y = \frac{R}{i\omega - a} \quad \text{and} \quad y = Ye^{i\omega t}. \tag{5}$$

The simplicity of the solution $y = Ye^{i\omega t}$ comes from one key fact: The derivative of $e^{i\omega t}$ is a multiple of $e^{i\omega t}$ (the multiplying factor is $i\omega$). This was not true for $\cos\omega t$. Its derivative brings in $\sin\omega t$. So we had to solve two real equations for M and N, while (5) is one complex equation for Y.

Complex Numbers : Rectangular and Polar

The complex number $z = x + iy$ has real part x and imaginary part y. The basic ideas are explained here ; more details are in Section 2.2. We plot all z in the **complex plane** (the real-imaginary plane). Figure 1.5 shows the particular number $z = 4 + 3i$ with $x = \text{Re } z = 4$ and $y = \text{Im } z = 3$. No problem with the rectangular form $4 + 3i$, except that multiplying and dividing are not at all convenient in $x - y$ coordinates.

The first figure also shows the **polar form** of the same number z. The magnitude (or modulus) is r. The phase is the angle θ. From x and y we can find r and θ.

The magnitude is $r = \sqrt{x^2 + y^2} = \sqrt{25} = 5$. *The angle* θ *has tangent* $y/x = 3/4$.

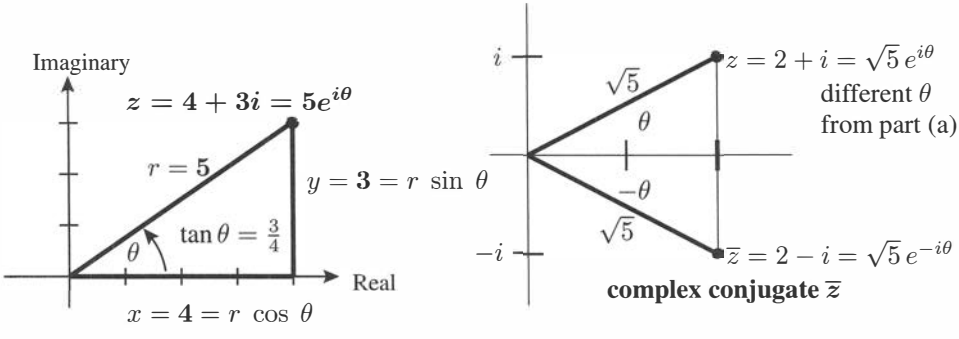


Figure 1.5: (a) $z = 4 + 3i$ is a point in the complex plane. Its polar form is $z = 5e^{i\theta}$.

The polar form is perfect for multiplication and division of complex numbers. To multiply $re^{i\theta}$ times $Re^{i\alpha}$, add the angles and multiply r times R. To divide, subtract the angles and divide r by R.

$$\textbf{Multiply} \quad (re^{i\theta})(Re^{i\alpha}) = \boldsymbol{rR\, e^{i(\theta+\alpha)}} \qquad \textbf{Divide} \quad \frac{re^{i\theta}}{Re^{i\alpha}} = \frac{\boldsymbol{r}}{\boldsymbol{R}}\, \boldsymbol{e^{i(\theta-\alpha)}} \tag{6}$$

The polar form is also perfect for squaring a complex number $re^{i\theta}$ and for $1/re^{i\theta}$:

$$\textbf{Square} \quad z^2 = (re^{i\theta})(re^{i\theta}) = \boldsymbol{r^2 e^{2i\theta}} \qquad \textbf{Invert} \quad \frac{1}{z} = \frac{1}{re^{i\theta}} = \frac{\boldsymbol{1}}{\boldsymbol{r}}\, \boldsymbol{e^{-i\theta}} \tag{7}$$

Let me compare that polar form of $1/z$ with $1/(x+iy)$. Multiply by $(x-iy)/(x-iy) = 1$.

$$\frac{1}{z} = \frac{1}{x+iy} = \frac{1}{x+iy}\frac{x-iy}{x-iy} = \frac{\boldsymbol{x - iy}}{\boldsymbol{x^2 + y^2}} \qquad \frac{1}{4+3i} = \frac{4-3i}{4^2+3^2} = \frac{1}{5}e^{-i\theta}$$

This number $x - iy$ appears often. It is the **complex conjugate** $\overline{z}$ of the number $z = x + iy$.

Notice that $x + iy$ times $x - iy$ is $x^2 + y^2$. In other words $\boldsymbol{z}$ **times** $\overline{\boldsymbol{z}}$ **is** $\boldsymbol{|z|^2 = r^2}$.

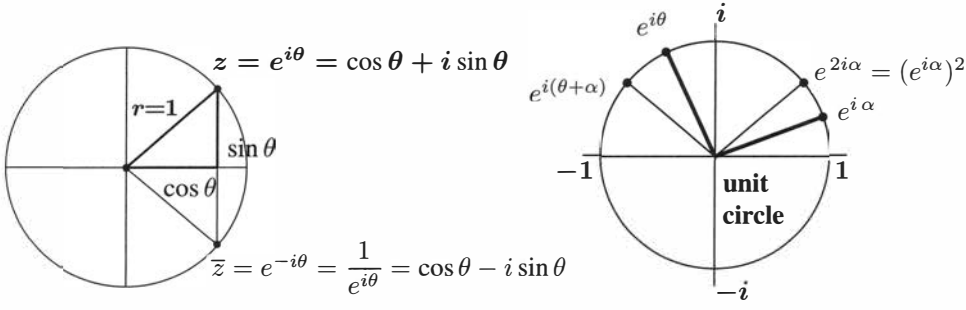


Figure 1.6: Points $e^{i\theta}$ on the unit circle have $r = 1$. When $e^{i\theta}$ multiplies $e^{i\alpha}$, angles add.

The Unit Circle

Figure 1.6 shows the **unit circle** , where every radial distance is $r = 1$. Then we just add the angles to multiply, or double the angles to square, or subtract the angles to divide:

On the circle $(e^{i\theta})(e^{i\alpha}) = e^{i(\theta+\alpha)}$ $(e^{i\theta})(e^{-i\theta}) = 1$ $\dfrac{1}{e^{i\theta}} = e^{-i\theta}$

$e^{-i\theta}$ is the complex conjugate of $e^{i\theta}$, the mirror image across the axis in Figure 1.6.

Example 1 Describe the paths of the numbers $\boldsymbol{e^{st}}$ and $\boldsymbol{e^{i\omega t}}$ and $\boldsymbol{e^{(s+i\omega)t}}$ in the complex plane (real s and real ω). The time t goes from 0 to ∞. Those paths start at 1.

Solution If $s > 0$, the number e^{st} goes from 1 out the real axis to infinity. If $s < 0$, then e^{st} goes from 1 in to zero. All real.

The path of $e^{i\omega t}$ goes around the unit circle with constant speed. At time $T = 2\pi/\omega$ (and also $2T$, $3T$, ...) it comes back to $e^{2\pi i} = 1$. The path goes clockwise if $\omega < 0$.

The path of $e^{(s+i\omega)t}$ **spirals outward** to infinity if $s > 0$. It spirals inward to zero if $s < 0$. At time $T = 2\pi/\omega$ it is a real number e^{sT}, because the factor $e^{i\omega T} = e^{2\pi i}$ is 1.

The Gain G and the Phase Lag α

The complex number $1/(i\omega - a)$ multiplies the input $q(t) = Re^{i\omega t}$ to give the output $y(t) = Ye^{i\omega t}$. **What is the magnitude of $1/(i\omega - \alpha)$ and what is its angle?** We need its polar form $\boldsymbol{1/(i\omega - a) = Ge^{-i\alpha}}$. Start with $i\omega - a = re^{i\alpha}$ and then invert:

$$\boldsymbol{i\omega - a = re^{i\alpha}} \qquad \boldsymbol{r = \sqrt{\omega^2 + a^2}} \quad \text{and} \quad \boldsymbol{\tan\alpha} = \frac{\text{imaginary part}}{\text{real part}} = -\frac{\boldsymbol{\omega}}{\boldsymbol{a}}.$$

We want $1/(re^{i\alpha})$. This will be $Ge^{-i\alpha}$. The gain is $\boldsymbol{G = 1/r = 1/\sqrt{\omega^2 + \alpha^2}}$:

Gain G
Phase angle α

$$\frac{1}{i\omega - a} = \frac{1}{r}e^{-i\alpha} = \frac{1}{\sqrt{\omega^2 + a^2}}e^{-i\alpha} = \boldsymbol{Ge^{-i\alpha}}. \tag{8}$$

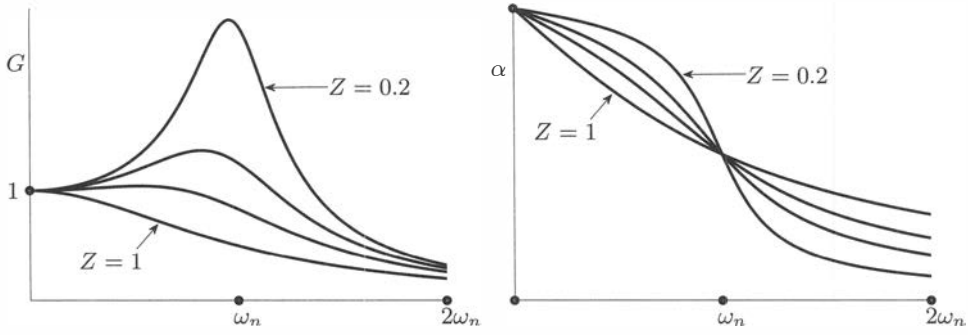


Figure 1.7: Dimensionless gain G and phase angle ϕ as functions of frequency ω.

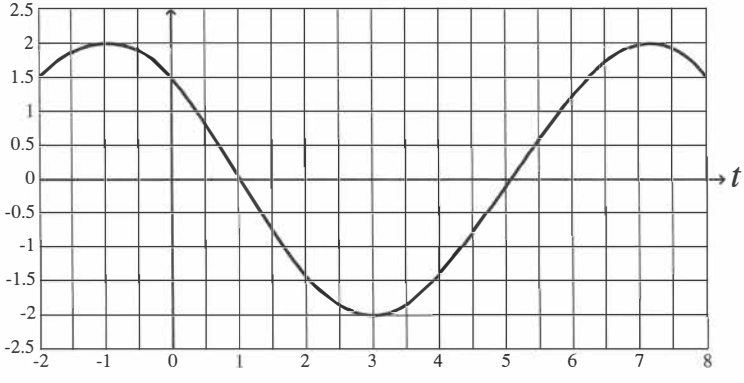
The gain $G(\omega)$ and the angle $\alpha(\omega)$ are often graphed. The graphs below are variations of "*Bode plots*." The amplitude response $G(\omega)$ is especially important, and you are very likely to see that gain G by itself—often including an extra factor $|a|$.

Note One common variation is to include the rate constant a in the forcing term $q(t) = a\,R\,e^{i\omega t}$. We still think of $R\,e^{i\omega t}$ as the input, then a gives q the right physical units. That factor a will appear in the output. So the gain $G = |\text{output}| \,/\, |\text{input}|$ will be increased by that factor $|a|$. *Then* $G = |a|/\sqrt{\omega^2 + a^2}$ is 1 *at the frequency* $\omega = 0$.

Sinusoids $R\cos(\omega t - \phi)$

The next page will show that any combination of $\cos\omega t$ and $\sin\omega t$ is a *shifted cosine*. It has frequency ω and amplitude R and phase lag ϕ. If you know ω and R and ϕ, it is no problem to graph $y(t) = \boldsymbol{R\cos(\omega t - \phi)}$. To go the other way, and *read off those three numbers from the graph*, is much more interesting.

This mystery sinusoid came from lecture notes for MIT's course 18.03. The website **mathlets.org** has interactive experiments. The question here is : **Find ω, R, and ϕ.**



The Sinusoidal Identity

We want to choose the magnitude R and the angle ϕ so that $A\cos\omega t + B\sin\omega t$ is the *real part* of $Re^{i(\omega t-\phi)}$. We can and will solve $y' - ay = Re^{i(\omega t-\phi)}$ quickly. When we take the real part of all terms in this differential equation, the correct input $q(t) = R\cos(\omega t - \phi)$ will appear on the right side and the correct output $y(t)$ will appear on the left side. The real equation will be solved in one step.

So we want this identity for the "sinusoidal" input $q(t)$:

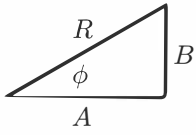
Sinusoidal identity $$\boldsymbol{A\cos\omega t + B\sin\omega t = R\cos(\omega t - \phi)} \tag{9}$$

The right side has the same period $2\pi/\omega$ as the left side—and only one term.

To find R and ϕ, expand $R\cos(\omega t - \phi)$ into $R\cos\omega t\cos\phi + R\sin\omega t\sin\phi$. Then match cosines to find A and match sines to find B:

$$A = R\cos\phi \quad \text{and} \quad B = R\sin\phi \qquad \boxed{\boldsymbol{A^2 + B^2 = R^2} \quad \text{and} \quad \boldsymbol{\tan\phi = \frac{B}{A}}.} \tag{10}$$

So we know $R = \sqrt{A^2 + B^2}$ and $\phi = \tan^{-1}(B/A)$ in the sinusoidal identity. The beauty of R and ϕ is that they match sinusoids to the polar form of complex numbers.



$A + iB$	$= Re^{i\phi}$	polar form of $A + iB$
R	$= \sqrt{A^2 + B^2}$	produces R and ϕ in the
$\tan\phi$	$= B/A$	sinusoidal identity (9)

For practice with this important formula, Problem 1 will develop a slightly different proof.

Example 2 Write $q(t) = \cos 3t + \sin 3t$ as $R\cos(3t - \phi)$: the real part of $Re^{i(3t-\phi)}$.

Solution $A = 1$ and $B = 1$ so that $R = \sqrt{2}$. The angle $\phi = \frac{\pi}{4}$ has $\tan\phi = B/A = 1$. Then $\cos 3t + \sin 3t = \boldsymbol{\sqrt{2}\cos(3t - \frac{\pi}{4})}$.

Example 3 Write the real part of $e^{i5t}/(\sqrt{3} + i)$ in the form $A\cos 5t + B\sin 5t$.

Solution $\sqrt{3} + i$ is $2e^{i\pi/6}$ (why?) Then $e^{i5t}/(\sqrt{3} + i)$ is $\frac{1}{2}e^{i(5t-\pi/6)}$. Its real part is

$$\frac{1}{2}\cos\left(5t - \frac{\pi}{6}\right) = \frac{1}{2}\left(\cos 5t\,\cos\frac{\pi}{6} + \sin 5t\,\sin\frac{\pi}{6}\right) = \frac{\sqrt{3}}{4}\cos 5t + \frac{1}{4}\sin 5t.$$

Real Solution y from Complex Solution y_c

The sinusoidal identity solves $y' - ay = A\cos\omega t + B\sin\omega t$ in three steps:

1. This equation is the real part of the complex equation $y_c' - ay_c = Re^{i(\omega t-\phi)}$.
2. The complex solution is $y_c = Re^{i(\omega t-\phi)}/(i\omega - a) = \boldsymbol{RGe^{i(\omega t-\phi-\alpha)}}$.
3. The real part of that complex solution y_c is the desired real solution $y(t)$.

Those three steps are **1** (real to complex) **2** (solve complex) **3** (complex to real). This will succeed. The second step expresses $1/(i\omega - \alpha)$ as $Ge^{-i\alpha}$ to keep the polar form. The third step produces $y = M\cos\omega t + N\sin\omega t$ directly as $\boldsymbol{y = RG\cos(\omega t - \phi - \alpha)}$.

Example 4 Take those three steps real-complex-real to solve $\boldsymbol{y' - y = \cos t - \sin t}$.

We have to find R, ϕ, G, and α from the numbers $a = 1$, $\omega = 1$, $A = 1$, and $B = -1$. Notice that $\boldsymbol{RG = 1}$.

$$R = \sqrt{A^2 + B^2} = \boldsymbol{\sqrt{2}} \quad \tan\phi = \frac{B}{A} = -1 \text{ and } \boldsymbol{\phi = -\frac{\pi}{4}} \quad G = \frac{1}{\sqrt{\omega^2 + a^2}} = \boldsymbol{\frac{1}{\sqrt{2}}}$$

The angle for $i\omega - a = i - 1$ is $\boldsymbol{\alpha = \frac{3\pi}{4}}$. Its tangent is $-\frac{\omega}{a} = -1$.

1. The sinusoidal identity is $\cos t - \sin t = \sqrt{2}\cos(t - \phi) = \boldsymbol{\sqrt{2}\cos(t + \pi/4)}$.
2. $\boldsymbol{y_{complex} = \dfrac{\sqrt{2}\,e^{i(t+\pi/4)}}{i - 1}}$. Here $\dfrac{1}{i\omega - a} = \dfrac{1}{i - 1} = Ge^{-i\alpha} = \dfrac{1}{\sqrt{2}}e^{-3\pi i/4}$.
3. $\boldsymbol{y_{complex} = RG\,e^{i(\omega t - \alpha - \phi)} = e^{i(t - \pi/2)}}$. Then $\boldsymbol{y_{real} = \cos\left(t - \dfrac{\pi}{2}\right) = \sin t}$.

That example was chosen so that $G = 1/\sqrt{2}$ cancelled $R = \sqrt{2}$. If we keep all the symbols R, ϕ, G, α then the solution $y_{real} = \boldsymbol{RG\cos(\omega t - \phi - \alpha)}$ from Step **3** must agree with the solution $y = M\cos\omega t + N\sin\omega t$ at the start of this section.

The key point in many applications is not necessarily the numbers in the formula for $y(t)$. Very often the goal is to see from the formula how $y(t)$ depends on parameters like a and ω in the differential equation. The gain $G = |\text{output}|/|\text{input}|$ is a convenient and very important guide.

The truth is that the complex solution is better. The sinusoidal identity shows how every combination $A\cos\omega t + B\sin\omega t$ is the real part $R\cos(\omega t - \phi)$ of a complex exponential $Re^{i(\omega t - \phi)}$. So we can convert real to complex and complex back to real.

In between, solve the complex form by using the ***frequency response*** $\boldsymbol{1/(i\omega - a)}$.

Conclusion When the input $q(t)$ is $Re^{i\omega t}$, the output $y(t)$ multiplies by $1/(i\omega - a)$. This multiplying factor is a complex number, and it changes with the frequency ω. We absolutely need to understand that number Y and graph its magnitude G and its phase.

■ REVIEW OF THE KEY IDEAS ■

1. (*Real*) $y' - ay = A\cos\omega t + B\sin\omega t$ leads to $\boldsymbol{y_{real} = M\cos\omega t + N\sin\omega t}$.
2. (*Sinusoidal identity*) $A\cos\omega t + B\sin\omega t$ equals $R\cos(\omega t - \phi)$ with $R^2 = A^2 + B^2$.
3. (*Complex*) $y' - ay = Re^{i(\omega t - \phi)}$ leads to $\boldsymbol{y_{complex} = Re^{i(\omega t - \phi)}/(i\omega - a)}$.
4. (*Complex gain*) $1/(i\omega - a) = \boldsymbol{Ge^{-i\alpha}}$ with $G = 1/\sqrt{\omega^2 + a^2}$ and $\tan\alpha = -\omega/a$.
5. (*Real part of the complex solution*) $\boldsymbol{y_{real} = \mathrm{Re}(y_{complex}) = RG\cos(\omega t - \alpha - \phi)}$.

Problem Set 1.5

Problems 1-6 are about the sinusoidal identity (9). It is stated again in Problem 1.

1 These steps lead again to the sinusoidal identity. This approach doesn't start with the usual formula $\cos(\omega t - \phi) = \cos\omega t \cos\phi + \sin\omega t \sin\phi$ from trigonometry. The identity says:

$$\textbf{If } A + iB = Re^{i\phi} \textbf{ then } A\cos\omega t + B\sin\omega t = R\cos(\omega t - \phi).$$

Here are the four steps to find that real part of $Re^{i(\omega t-\phi)}$. Explain $A - iB$ in Step 3.

$$R\cos(\omega t - \phi) = \text{Re}\left[Re^{i(\omega t-\phi)}\right] = \text{Re}\left[e^{i\omega t}(Re^{-i\phi})\right] = \text{(what is } Re^{-i\phi}\text{?)}$$

$$= \text{Re}\left[(\cos\omega t + i\sin\omega t)(A - iB)\right] = A\cos\omega t + B\sin\omega t.$$

2 To express $\sin 5t + \cos 5t$ as $R\cos(\omega t - \phi)$, what are R and ϕ?

3 To express $6\cos 2t + 8\sin 2t$ as $R\cos(2t - \phi)$, what are R and $\tan\phi$ and ϕ?

4 Integrate $\cos\omega t$ to find $(\sin\omega t)/\omega$ in this complex way.

(i) $dy_{\text{real}}/dt = \cos\omega t$ is the real part of $dy_{\text{complex}}/dt = e^{i\omega t}$.

(ii) Take the real part of the complex solution.

5 The sinusoidal identity for $A = 0$ and $B = -1$ says that $-\sin\omega t = R\cos(\omega t - \phi)$. Find R and ϕ.

6 Why is the sinusoidal identity useless for the source $q(t) = \cos t + \sin 2t$?

7 Write $2+3i$ as $re^{i\phi}$, so that $\frac{1}{2+3i} = \frac{1}{r}e^{-i\phi}$. Then write $y = e^{i\omega t}/(2+3i)$ in polar form. Then find the real and imaginary parts of y. And also find those real and imaginary parts directly from $(2-3i)e^{i\omega t}/(2-3i)(2+3i)$.

8 Write these functions $A\cos\omega t + B\sin\omega t$ in the form $R\cos(\omega t - \phi)$: Right triangle with sides A, B, R and angle ϕ.

1) $\cos 3t - \sin 3t$ 2) $\sqrt{3}\cos\pi t - \sin\pi t$ 3) $3\cos(t-\phi) + 4\sin(t-\phi)$

Problems 9-15 solve real equations using the real formula (3) for M and N.

9 Solve $dy/dt = 2y + 3\cos t + 4\sin t$ after recognizing a and ω. Null solutions Ce^{2t}.

10 Find a particular solution to $dy/dt = -y - \cos 2t$.

11 What equation $y' - ay = A\cos\omega t + B\sin\omega t$ is solved by $y = 3\cos 2t + 4\sin 2t$?

12 The particular solution to $y' = y + \cos t$ in Section 1.4 is $y_p = e^t \int e^{-s}\cos s\, ds$. Look this up or integrate by parts, from $s = 0$ to t. Compare this y_p to formula (3).

13 Find a solution $y = M\cos\omega t + N\sin\omega t$ to $y' - 4y = \cos 3t + \sin 3t$.

14 Find the solution to $y' - ay = A\cos\omega t + B\sin\omega t$ **starting from $y(0) = 0$.**

15 If $a = 0$ show that M and N in equation (3) still solve $y' = A\cos\omega t + B\sin\omega t$.

Problems 16-20 solve the complex equation $y' - ay = Re^{i(\omega t - \phi)}$.

16 Write down complex solutions $y_p = Ye^{i\omega t}$ to these three equations:

(a) $y' - 3y = 5e^{2it}$ (b) $y' = Re^{i(\omega t-\phi)}$ (c) $y' = 2y - e^{it}$

17 Find complex solutions $z_p = Ze^{i\omega t}$ to these complex equations:

(a) $z' + 4z = e^{8it}$ (b) $z' + 4iz = e^{8it}$ (c) $z' + 4iz = e^{8t}$

18 Start with the real equation $y' - ay = R\cos(\omega t - \phi)$. Change to the complex equation $z' - az = Re^{i(\omega t-\phi)}$. Solve for $z(t)$. Then take its real part $y_p = \text{Re}\, z$.

19 What is the initial value $y_p(0)$ of the particular solution y_p from Problem 18? If the desired initial value is $y(0)$, how much of the null solution $y_n = Ce^{at}$ would you add to y_p?

20 Find the real solution to $y' - 2y = \cos\omega t$ starting from $y(0) = 0$, in three steps: Solve the complex equation $z' - 2z = e^{i\omega t}$, take $y_p = \text{Re}\, z$, and add the null solution $y_n = Ce^{2t}$ with the right C.

Problems 21-27 solve real equations by making them complex. First a note on α.

Example 4 was $y' - y = \cos t - \sin t$, with growth rate $a = 1$ and frequency $\omega = 1$. The magnitude of $i\omega - a$ is $\sqrt{2}$ and the polar angle has $\tan\alpha = -\omega/a = -1$. Notice: *Both $\alpha = 3\pi/4$ and $\alpha = -\pi/4$ have that tangent*! How to choose the correct angle α?

The complex number $i\omega - a = i - 1$ is in the *second quadrant*. Its angle is $\alpha = 3\pi/4$. **We had to look at the actual number and not just the tangent of its angle**.

21 Find r and α to write each $i\omega - a$ as $re^{i\alpha}$. Then write $1/re^{i\alpha}$ as $Ge^{-i\alpha}$.

(a) $\sqrt{3}\,i + 1$ (b) $\sqrt{3}\,i - 1$ (c) $i - \sqrt{3}$

22 Use G and α from Problem 21 to solve (a)-(b)-(c). Then take the real part of each equation and the real part of each solution.

(a) $y' + y = e^{i\sqrt{3}t}$ (b) $y' - y = e^{i\sqrt{3}t}$ (c) $y' - \sqrt{3}y = e^{it}$

23 Solve $y' - y = \cos \omega t + \sin \omega t$ in three steps: real to complex, solve complex, take real part. This is an important example.

(1) Find R and ϕ in the sinusoidal identity to write $\cos \omega t + \sin \omega t$ as the real part of $Re^{i(\omega t - \phi)}$.

(2) Solve $y' - y = e^{i\omega t}$ by $y = Ge^{-i\alpha}e^{i\omega t}$. Multiply by $Re^{-i\phi}$ to solve $z' - z = Re^{i(\omega t - \phi)}$.

(3) Take the real part $y(t) = \text{Re}\, z(t)$. Check that $y' - y = \cos \omega t + \sin \omega t$.

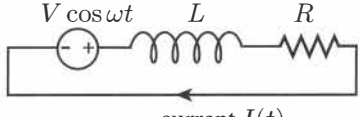
24 Solve $y' - \sqrt{3}y = \cos t + \sin t$ by the same three steps with $a = \sqrt{3}$ and $\omega = 1$.

25 (**Challenge**) Solve $y' - ay = A \cos \omega t + B \sin \omega t$ in two ways. First, find R and ϕ on the right and G and α on the left. Show that the final real solution $RG \cos(\omega t - \phi - \alpha)$ agrees with $M \cos \omega t + N \sin \omega t$ in equation (2).

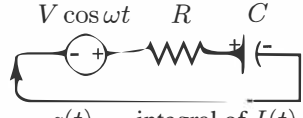
26 We don't have resonance for $y' - ay = Re^{i\omega t}$ when a and $\omega \neq 0$ are real. *Why not*? (Resonance appears when $y_n = Ce^{at}$ and $y_p = Ye^{ct}$ share the exponent $a = c$.)

27 If you took the imaginary part $y = \text{Im}\, z$ of the complex solution to $z' - az = Re^{i(\omega t - \phi)}$, what equation would $y(t)$ solve? Answer first with $\phi = 0$.

Problems 28-31 solve first order circuit equations: not RLC but RL and RC.



current $I(t)$



$q(t) =$ integral of $I(t)$

28 Solve $\boldsymbol{L\, dI/dt + RI(t) = V \cos \omega t}$ for the current $I(t) = I_n + I_p$ in the RL loop.

29 With $L = 0$ and $\omega = 0$, that equation is Ohm's Law $V = IR$ for direct current. The **complex impedance** $Z = R + i\omega L$ replaces R when $L \neq 0$ and $I(t) = Ie^{i\omega t}$.

$$L\, dI/dt + RI(t) = (\boldsymbol{i\omega L + R})\boldsymbol{Ie^{i\omega t} = Ve^{i\omega t}} \quad \text{gives} \quad \boldsymbol{ZI = V}.$$

What is the magnitude $|Z| = |R + i\omega L|$? What is the phase angle in $Z = |Z|e^{i\theta}$? Is the current $|I|$ larger or smaller because of L?

30 Solve $\boldsymbol{R\dfrac{dq}{dt} + \dfrac{1}{C}q(t) = V \cos \omega t}$ for the charge $q(t) = q_n + q_p$ in the RC loop.

31 Why is the complex impedance now $Z = R + \frac{1}{i\omega C}$? Find its magnitude $|Z|$. **Note that mathematics prefers $i = \sqrt{-1}$, we are not conceding yet to $j = \sqrt{-1}$!**